MCQ Questions on chapter 1&2&3

1. The number of pixels stored in the frame buffer of a graphics system is known as
a) Resolution
b) Depth
c) All of the mentioned
d) Neither a nor b

2. In graphical system, the array of pixels in the picture are stored in
a) Memory
b) Frame buffer
c) Processor
d) All of the mentioned

3. The maximum number of points that can be displayed without overlap on a CRT is referred as
a) Picture
b) Resolution
c) Persistence
d) Neither b nor c

4. In which system, the Shadow mask methods are commonly used in
a) Raster-scan system
b) Random-scan system
c) Only b
d) Both a and b

5. The process of digitizing a given picture definition into a set of pixel-intensity for storage in the frame buffer is called
a) Rasterization
b) Encoding
c) Scan conversion
d) True color system

6. Random-scan system mainly designed for
a) Realistic shaded screen
b) Fog effect
c) Line-drawing applications
d) Only b

7. The primary output device in a graphics system is_________
a) Scanner
b) Video monitor
c) Neither a nor b
d) Printer

8. On raster system, lines are plotted with
a) Lines
b) Dots
c) Pixels
d) None of the mentioned

9. Expansion of line DDA algorithm is
a) Digital difference analyzer
b) Direct differential analyzer
c) Digital differential analyzer
d) Data differential analyzer

10. An accurate and efficient raster line-generating algorithm is
a) DDA algorithm
b) Mid-point algorithm
c) Parallel line algorithm
d) Bresenham's line algorithm

11. In Bresenham's line algorithm, if the distances d1 < d2 then decision parameter Pk is______
a) Positive
b) Equal
c) Negative
d) Option a or c

12. To apply the midpoint method, we define
a) F_circle(x, y)= $x^2 + y^2 - r^2$
b) F_circle(x, y)= $x + y^2 - r^2$
c) F_circle(x, y)= $x^2 - y^2 - r^2$
d) F_circle(x, y)= $x^2 + y^2 - z^2$

13. Refresh rate is
A. The rate at which the number of bit planes are accessed at a given time
B. The rate at which the picture is redrawn
C. The frequency at which the aliasing takes place
D. The frequency at which the contents of the frame buffer is sent to the display monitor

14. Frame buffer is
A. The memory area in which the image, being displayed, is stored
B. The device which controls the refresh rate
C. The device used for displaying the colors of an image

D. The memory area in which the graphics package is stored

15. In Bresenham's line algorithm, if the distances d1 < d2 then decision parameter Pk is______ مكرر!
A) Positive
B) Equal
C) Negative
D) Option a or c

16. In graphical system, the array of pixels in the picture are stored in مكرر!
A) Memory
B) Frame buffer
C) Processor
D) All of the mentioned

17. The maximum number of points that can be displayed without overlap on a CRT is referred as مكرر!
A) Picture
B) Resolution
C) Persistence
D) Neither b nor c

18. In which system, the Shadow mask methods are commonly used in
A) Raster-scan system
B) Random-scan system
C) Only b
D) Both a and b

19. The process of digitizing a given picture definition into a set of pixel-intensity for storage in the frame buffer is called
A) Rasterization
B) Encoding
C) Scan conversion
D) True color system

20. Random-scan system mainly designed for
A) Realistic shaded screen
B) Fog effect
C) Line-drawing applications
D) Only b

21. The primary output device in a graphics system is_________
A) Scanner
B) Video monitor

C) Neither a nor b
D) Printer

22. On raster system, lines are plotted with
A) Lines
B) Dots
C) Pixels
D) None of the mentioned

23. Refresh rate is
A) The rate at which the number of bit planes are accessed at a given time
B) The rate at which the picture is redrawn
C) The frequency at which the aliasing takes place
D) The frequency at which the contents of the frame buffer is sent to the display monitor

24. Frame buffer is
A) The memory area in which the image, being displayed, is stored
B) The device which controls the refresh rate
C) The device used for displaying the colors of an image
D) The memory area in which the graphics package is stored

25. What does LCD stand for?
A) Liquid Crystal Display
B) Light Crystal Display
C) Liquid Crystal Diode
D) Light Crystal Diode

26. Which of the following is a characteristic of LCDs?
A) They use a cathode ray tube (CRT)
B) They use a backlight
C) They use a plasma panel
D) They use a laser projector

27. What is the purpose of the polarizer in an LCD?
A) To block or allow light to pass through the LCD
B) To control the voltage applied to the LCD
C) To generate the images displayed on the LCD
D) To connect the LCD to a computer

28. Which component is responsible for generating the images displayed on an LCD?
A) Backlight
B) Polarizer
C) Liquid crystal layer
D) Controller

29. Which of the following is a common application of LCDs?
A) TVs
B) Computer monitors
C) Mobile phones
D) All of the above

30. Which type of flat display uses a layer of organic material to produce images?
A) TN (Twisted NematiC) LCD
B) IPS (In-Plane Switching) LCD
C) VA (Vertical Alignment) LCD
D) OLED (Organic Light-Emitting Diode) LCD

31. Which industry uses LCDs in devices such as medical imaging equipment and patient monitoring systems?
A) Consumer electronics
B) Medical devices
C) Industrial automation
D) Aerospace

32. What does OLED stand for?
A) Organic Light-Emitting Diode
B) Optical Light-Emitting Diode
C) Organic Laser-Emitting Diode
D) Optical Laser-Emitting Diode

33. Which of the following is a characteristic of OLED displays?
A) They use a backlight
B) They use a liquid crystal layer
C) They emit their own light
D) They require a separate light source

34. Which of the following is an advantage of OLED displays over LCDs?
A) Higher power consumption
B) Lower contrast ratio
C) Faster response time
D) Lower viewing angles

35. Which of the following benefits do OLED displays provide?
A) Improved color accuracy
B) Increased brightness
C) Reduced power consumption
D) All of the above

36. Which industry uses OLED displays in devices such as smartphones and smartwatches?
A) Consumer electronics
B) Automotive
C) Aerospace
D) Medical devices

36. What does LED stand for?
A) Light Emitting Diode
B) Liquid Emitting Diode
C) Laser Emitting Diode
D) Luminescent Emitting Diode

37. Which of the following is an advantage of LED displays?
A) Higher power consumption
B) Lower brightness
C) Faster response time
D) Lower viewing angles

38. Which of the following benefits do LED displays provide? مكرر!
A) Improved color accuracy
B) Increased contrast ratio
C) Reduced power consumption
D) All of the above

39. Which industry uses LED displays in devices such as TVs and monitors?
A) Consumer electronics
B) Automotive
C) Aerospace
D) Medical devices

40. What is the primary technology used in plasma displays?
A) Liquid crystal display (LCD)
B) Light-emitting diode (LED)
C) Plasma display panel (PDP)
D) Cathode ray tube (CRT)

41. Which of the following is a characteristic of plasma displays?
A) They use a backlight
B) They use a liquid crystal layer
C) They emit their own light
D) They require a separate light source

42. Which of the following is a component of a plasma display panel?
A) Electron gun
B) Phosphor coating
C) Address electrodes
D) Backlight

الأتنين صح !

43. Which of the following is an advantage of plasma displays? مكرر!
A) Higher power consumption
B) Lower contrast ratio
C) Wider viewing angles
D) Lower brightness

الأتنين صح !

44. Which industry uses plasma displays in devices such as TVs and monitors?
A) Consumer electronics
B) Automotive
C) Aerospace
D) Medical devices

45. What is the process by which plasma displays produce images?
A) Electroluminescence
B) Photoluminescence
C) Cathodoluminescence
D) Electromagnetic induction

46. Which of the following gases is commonly used in plasma displays?
A) Neon
B) Xenon
C) Argon
D) All of the above

47. Which of the following is a characteristic of plasma displays? مكرر!
A) High power consumption
B) Low contrast ratio
C) Wide viewing angles
D) Low brightness

الأتنين صح !

48. What is a disadvantage of plasma displays?
A) High power consumption
B) Low contrast ratio
C) Limited viewing angles
D) All of the above

49. What is computer graphics?
A) The study of computer hardware
B) The study of computer software
C) The study of creating images using computers
D) The study of computer networking

50. What is the primary goal of computer graphics?
A) To create realistic images
B) To create interactive images
C) To create images that can be manipulated
D) All of the above

51. What is the difference between raster graphics and vector graphics?
A) Raster graphics use pixels, while vector graphics use lines and curves
B) Raster graphics use lines and curves, while vector graphics use pixels
C) Raster graphics are used for 2D images, while vector graphics are used for 3D images
D) Raster graphics are used for 3D images, while vector graphics are used for 2D images

52. What is the purpose of a graphics pipeline?
A) To render 2D images
B) To render 3D images
C) To transform 3D models into 2D images
D) To perform tasks such as clipping and culling

53. What is the purpose of a framebuffer?
A) To store the final rendered image
B) To store the 3D model data
C) To store the vertex data
D) To store the pixel data

54. What is the purpose of the rasterizer?
A) To transform 3D vertices into 2D screen coordinates
B) To perform lighting calculations
C) To perform texture mapping
D) To convert 3D geometry into 2D pixels

55. What is the purpose of the framebuffer? مكرر!
A) To store the final rendered image
B) To store the 3D model data
C) To store the vertex data
D) To store the pixel data

56. What is the difference between a 2D point and a 3D point?
A) A 2D point has two coordinates, while a 3D point has three coordinates
B) A 2D point has three coordinates, while a 3D point has two coordinates
C) A 2D point is used for 3D graphics, while a 3D point is used for 2D graphics
D) A 2D point is used for 2D graphics, while a 3D point is used for 3D graphics

57. What is the purpose of the Bresenham's line algorithm?
A) To draw lines on a raster display
B) To perform lighting calculations
C) To transform 3D points into 2D points
D) To add surface detail to 3D objects

58. What is the difference between the DDA algorithm and the Bresenham's line algorithm?
A) The DDA algorithm uses floating-point arithmetic, while the Bresenham's line algorithm uses integer arithmetic
B) The DDA algorithm uses integer arithmetic, while the Bresenham's line algorithm uses floating-point arithmetic
C) The DDA algorithm is used for 2D graphics, while the Bresenham's line algorithm is used for 3D graphics
D) None of the mentioned